Date

Monday
26/10/26.

LECTURE - 17

Divide & Conquer

- Find an optimal solution since we traversed every possible sub problem. But here we've overlapping subproblems.
- Solution space is graph

Dynamic programic:

- Store solutions to eliminate overlapping. It finds the optimal solution
- Solution space is tree

Technique.

Greedy Algorithm: We always use our brains & time to select Best OPTION

- This is like the forwarding router in CN kay current node pr aa kr we'll decide kay which is the next best solution towards the goal; baaki saray paths are ignored.
- We can't come back to change the branch of sub problem.
- Provides sub-optimal solution since "har step pr best option" might not ultimate give the best path from start → goal.
- Mostly faster than other two
- Might not be able to complete the traversal to provide sol. sometimes
- Solution space is List.

Fractional Knapsack:

Here we can choose any fractional part of item as well instead of entire item to maximise the profit. But DP fails in fractional knapsack as for DP we need to make an array. Whereas here we've infinite possibilities bw 0 → 1 & so on in decimals so we can't create infinite size array. THEREFORE we use GREEDY ALGO.

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item(i)	1	2	3	4	5	6
Price(i) $	20	30	32	10	3	4
Weight (i) kg	2	5	7	3	1	2
Pi/wi	$\frac{20}{2}=10$	$\frac{30}{5}=6$	4.5	3.3	3	2

W = 9 (space of sack)

OPTIONS IN GREEDY :

(1) Pick an item with maximum profit wise:

We'll start with item 2 with max price (30) then choose either the entire item to add in sack or a part/fraction of that item DEPENDING on the space left in sack (W=9)

$S = \{$ item number, Total/fractional price, Total/fractional weight $\}$

$S_1 = \{$ 3rd, 2nd $\}$ → $\{$ $32, $12 $\}$ → $\{$ 7kg, 2kg $\}$

- 3rd: complete price & weight of 3rd
- 2nd: $\frac{2}{5} \times 30$ → total weight of 2nd (weight used)
- 9 − 7 = 2 → Weight of sack

$S_1 = 32 + 12$

$= 44$ {3

* we started from 3rd item in S1 since it had max profit. Then 2nd highest profit was of 2nd item.

(2) Pick an item with least weight wise so more items:

We'll start with lowest weight i.e: item 5 and then move on to a bit higher weight i.e: item 1st OR 6th

$S_2 = \{$ 5th, 1st, 6th, 4th, 2nd $\}$

$\{$ $3 + $20 + $4 + $10 + $6 $\}$

$\{$ 1kg, 2kg, 2kg, 3kg, 1kg $\}$

(1kg, 2kg, 2kg, 3kg: complete; 1kg: → 1/5th)

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$S_2 = 3 + 20 + 4 + 10 + 6$

$= \boxed{43}$

③ P_i/w_i → Commute the price/weight ratio & Pick Maximum:

we'll mix weight with price since we don't know either ^{max}price is better or ^{min}weight.

Therefore we'll commute P_i/w_i in $O(n)$ & then either SORT 'P_i/w_i' in $O(n \log n)$ or use MAX HEAP: $O(n \log n)$

* Make a row of 'P_i/w_i' in table of question.

	1st	2nd	3rd	
S_3 =	$20	+ $30	+ $9	→ 4.5 × 2 = 9
10 × 2 ←	2kg	5kg	2kg	→ 9 − 5 − 2 = 2 (remaining weight)

since 1st item have the max P_i/w_i (i.e: 10) therefore we'll grab most of 1st items, then move to next max 'P_i/w_i' ie: item 2 with ($P_i/w_i = 6$)

$S_3 = 20 + 30 + 9 = \boxed{59}$ $\quad \overset{1^{st}}{\{10, 10,} \overset{2nd}{6, 6, 6, 6, 6,} \overset{3rd}{4.5, 4.5}\}$

- Now we can see out of S1 (42), S2 (43) & S3 (59) S3 has maximum profit.

⇒ How to see if the solution of Greedy is Optimal or Sub optimal

- For getting vaazi with sub optimal solution we'll use proofs by contradiction and take one new small data set which will beat the ^{max}profit (i.e: 59) with the same technique/option #3 (i.e: price/weight ratio).

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- We can proof that our provided solution is Optimal by replacing any value in the set with the value that didn't got choosen from the table.

e.g: $S_3 = \{10, 10, 6, 6, 6, 6, 4.5, 4.5\}$

- we can replace this 4.5 pi/wi value with value of another all like '3.3' which did not get selected.
- But here we can see that total profit would decrease if we'll choose 3.3.

Therefore we'll try other options not choose too & then finally declare S_3 as the optimal solution.

- Greedy algo comes with this proof so we know where we stand.

⇒ Same problem from 0/1 (non fraction) knapsack:

We can choose any technique (1,2,3) from here to solve 0/1 & method would be same but lesser profit as space is left too.